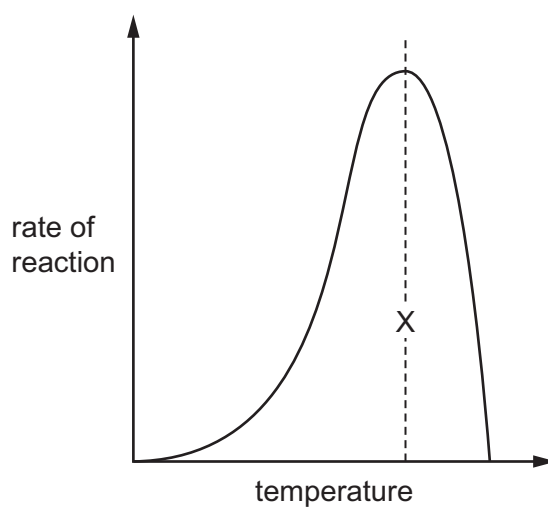


- 16** The graph shows the effect of temperature on the rate at which the enzyme in a biological washing powder digests and removes fruit juice stains.



Which statements explain the shape of the graph at temperatures higher than X?

- 1 Bonds are broken between the R groups of the amino acids in the polypeptide chains of the enzyme.
- 2 There are more collisions between the enzyme and its substrate.
- 3 The tertiary structure of the enzyme is altered.
- 4 The shapes of the active site and the substrate are no longer complementary.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4